

Avalon-MM SD Card Controller IP Core

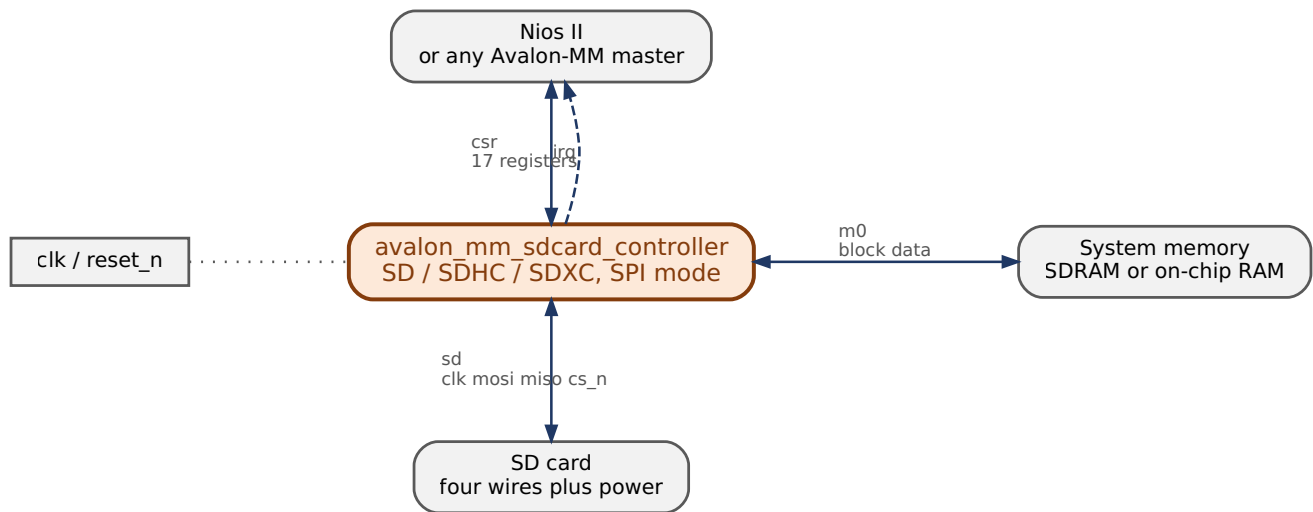
Block Diagrams

IP core	<code>altera_avalon_mm_sdcard_controller</code>
Core version	1.0
Document version	1.1
Date	September 2026
Target	Intel FPGA / Quartus Prime, Platform Designer

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1. What it connects to



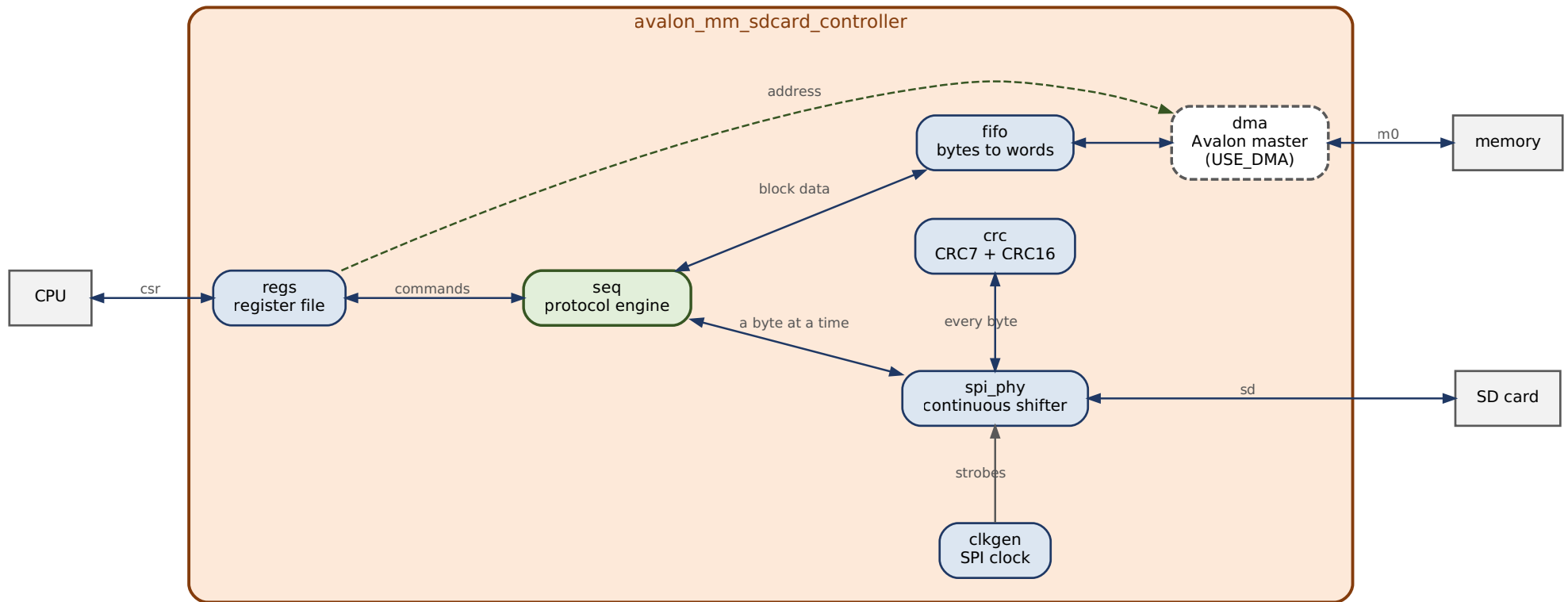
Interface	Type	Carries
csr	Avalon-MM slave	The 17 registers software drives
m0	Avalon-MM master, bursting	Block data, to and from memory
irq	Interrupt sender	One line
sd	Conduit	clk, mosi, miso, cs_n — plus cd_n and wp_n if USE_CARD_DETECT

Two Avalon-MM interfaces, not one. **csr** is how software drives the core. **m0** is how the core moves block data by itself.

That second interface is the difference between one command per block and 256 memory accesses per block. A 512-byte block is 128 words each way; without a master, the CPU has to move all of them, and it cannot look away while it does — fall behind and the shifter starves.

m0 is optional. Set `USE_DMA = 0` and the interface disappears from the component; software then moves words through the **DATA** register instead. That configuration is one of the five the regression sweeps, so it is tested rather than assumed.

2. Inside the core



Nine RTL files, 3,127 lines.

Module	Job
pkg	Register map, bit positions, protocol constants
regs	The <code>csr</code> slave. Read latency 1, one interrupt mask
seq	The protocol engine — 20 states
spi_phy	The shifter. Runs continuously, keeps one byte prefetched
clkgen	Divides <code>clk</code> to the SPI rate
crc	CRC7 on commands, CRC16 on data
fifo	Word-wide buffer with a byte packer on the card side
dma	The <code>m0</code> master

The core forks at the sequencer. Block data goes out through the FIFO and the master; single bytes — commands, responses, tokens — go out through the shifter.

2.1 The shifter does not stop

It keeps one byte prefetched, so the next byte is queued before the current one finishes and the SPI clock never pauses between bytes.

That is worth **16,696 SPI clocks to move 2,048 bytes**, against a floor of 16,384 — **98.1% of line rate**. The regression asserts that cycle count, so a change that reintroduces an idle clock fails the build rather than quietly costing 11%.

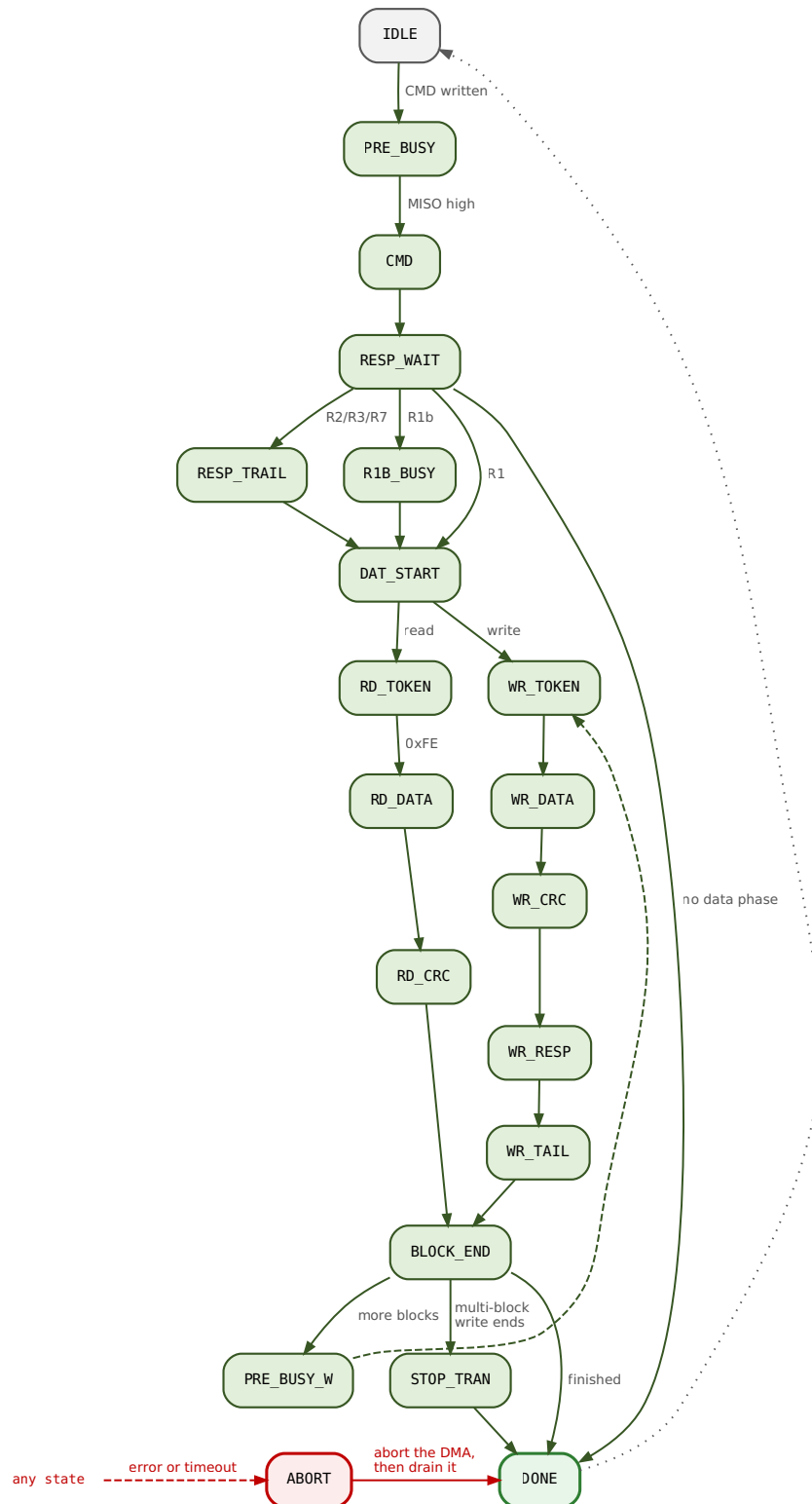
It also decides how the CRCs are built. A bit-serial CRC needs eight cycles per byte, which only exist if the shifter pauses. This one does not pause, so both CRCs take a whole byte at a time.

2.2 The FIFO is word-wide, with a packer

Its two ends run at different widths and different rates: the shifter moves one byte per eight SPI clocks, the master moves four bytes per host cycle.

At the default `FIFO_DEPTH_BYTES = 1024` it holds two blocks, so the DMA can drain one while the shifter fills the next. At 512 it holds one, the overlap disappears, and the data path has to refill mid-transfer — which is why 512 is in the sweep.

3. How a transfer runs



Read it top to bottom. Every operation is a command; some commands then have a data phase, which is either the read chain or the write chain, and both rejoin at **BLOCK_END**.

Error exits are not drawn. Every state has one, and they all go to **ABORT**.

Three states are worth naming, because their purpose is not in their names:

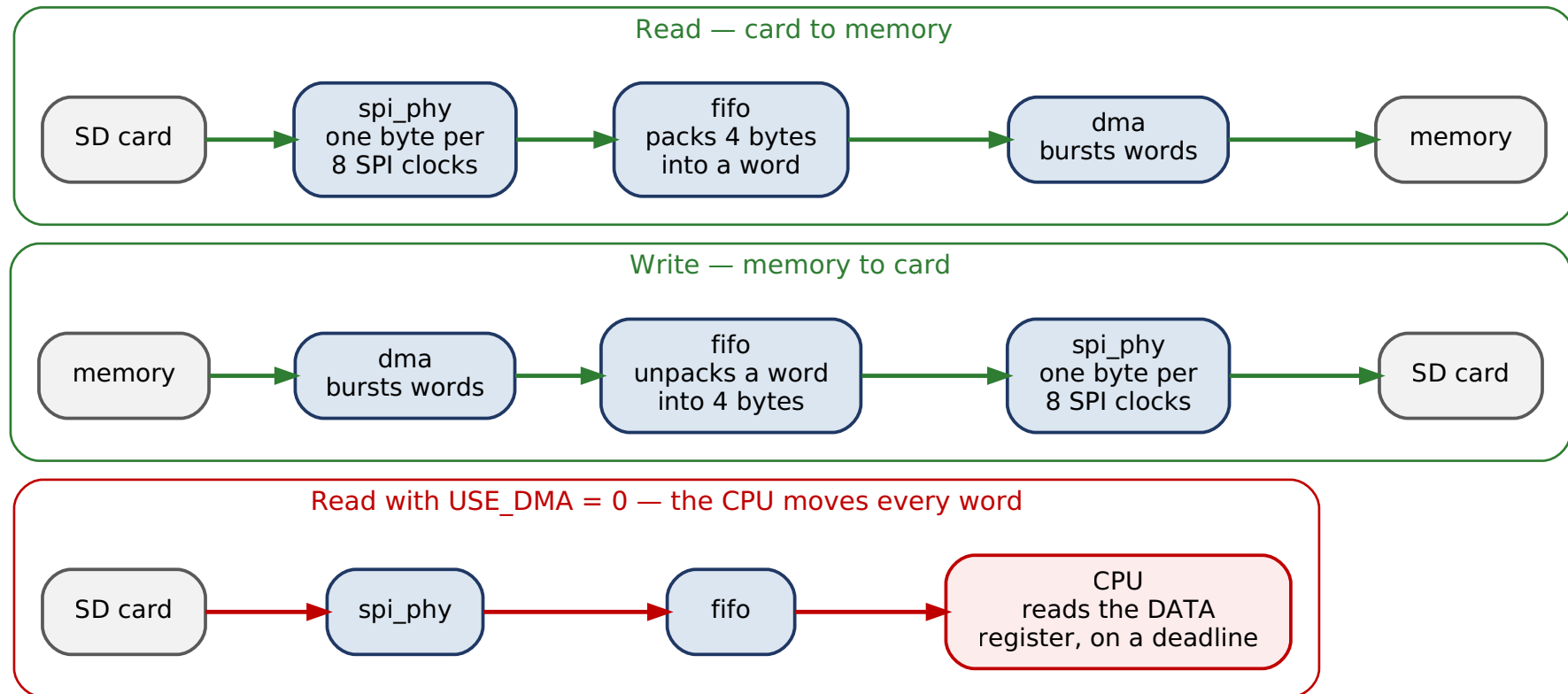
PRE_BUSY clocks **0xFF** with **CS** high before every command. A card still programming an earlier block holds **MISO** low, and this is how the core finds out. It also means the shifter is always already running when a sending state begins.

WR_TAIL covers the eight clocks after a data-response token, during which busy on **MISO** does not yet mean anything. Sample earlier and a card that has not started looks like a card that has finished.

R1B_BUSY is only for R1b commands, where the card holds the line until the operation completes.

ABORT does not go straight to **IDLE**. It goes through **DONE**, so the DMA is told to abort and then allowed to drain. An Avalon master that stops issuing beats part-way through a burst hangs the interconnect — so giving up on a transfer takes more care than never starting one.

4. Where the bytes go



Read and write are mirror images. The third row is the same core with `USE_DMA = 0`.

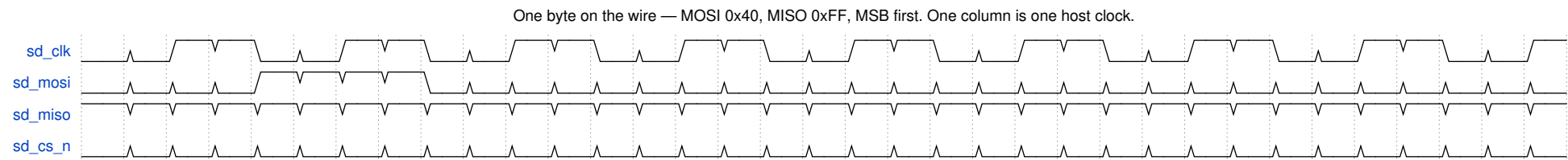
In that third case the CPU is on a deadline: nothing else is keeping the buffer moving, so software that is slow to service the `DATA` register starves the shifter and the data-phase timeout fires.

That timeout exists because of a bug. `RD_DATA` and `WR_DATA` originally had no timeout at all, so a starved data phase hung the core until a soft reset. With a master attached it cannot happen, which is why it survived until the PIO configuration was first run. The bound is on time **without progress**, so a slow block never trips it and a stalled one does.

5. On the wire

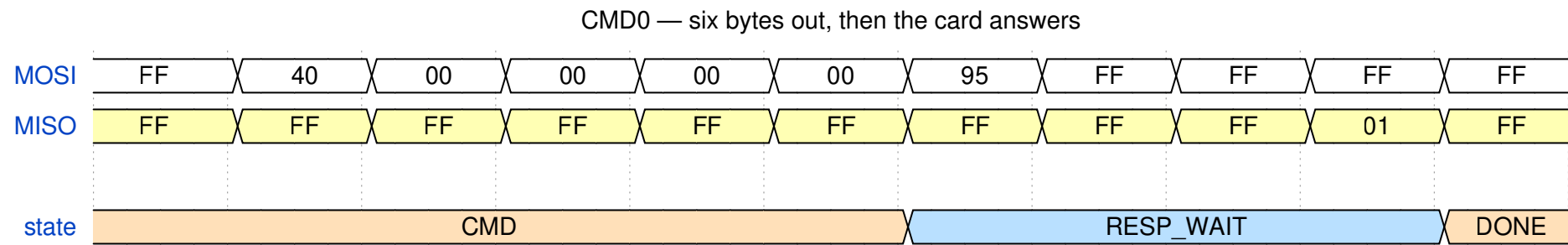
Five figures, all cut from one recorded simulation. The bytes, tokens, CRC values and state names in them were produced by the RTL in this repository.

5.1 One byte, at bit level



CPOL = 0, CPHA = 0. `MOSI` changes on the falling edge of `sd_clk`; both ends sample on the rising edge. This is the one thing a byte-level view cannot show, and the one to get right when wiring to a real card.

5.2 A command and its response

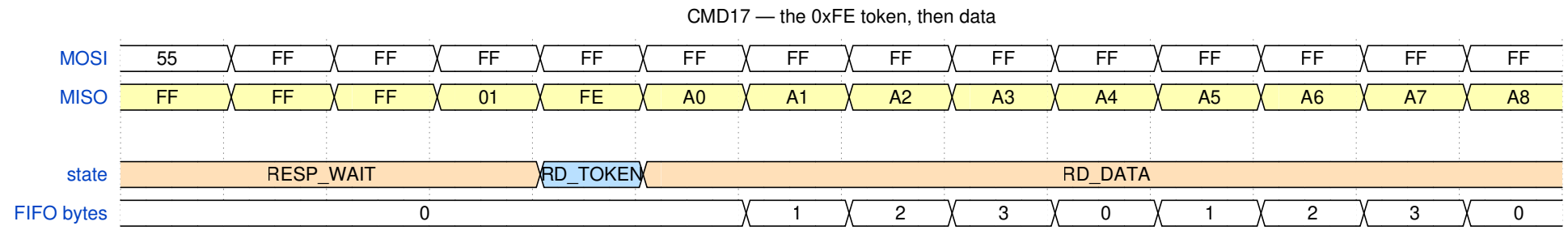


Six bytes out: `0x40` | `index`, four argument bytes, then CRC7 shifted up one with a stop bit in bit 0. Then the host clocks `0xFF` until the card answers.

`N_CR` — the gap before the response — is a **range**, 0 to 8 byte-times, not a fixed latency. So the core polls for a byte with bit 7 clear rather than waiting a set time. Here the card took two.

`0x95` is the CRC7 of CMD0 with a zero argument, which is why that constant appears in every SD initialisation routine ever written.

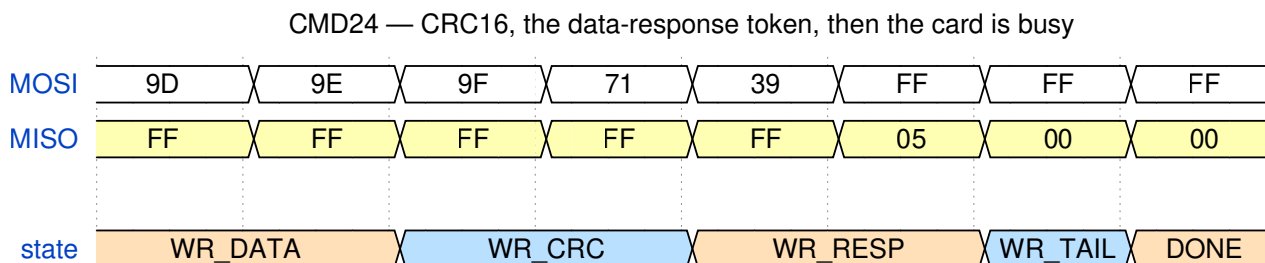
5.3 The start of a block read



After the R1 the card may take as long as it likes, so the core sits in `RD_TOKEN` clocking `0xFF`. `0xFE` starts the block. Anything with the top four bits clear is a data error token instead, and lands in `ERR_INFO` rather than being treated as data.

Watch the FIFO count: it rises as bytes arrive and falls as the DMA drains them, rather than climbing to 512 and then emptying.

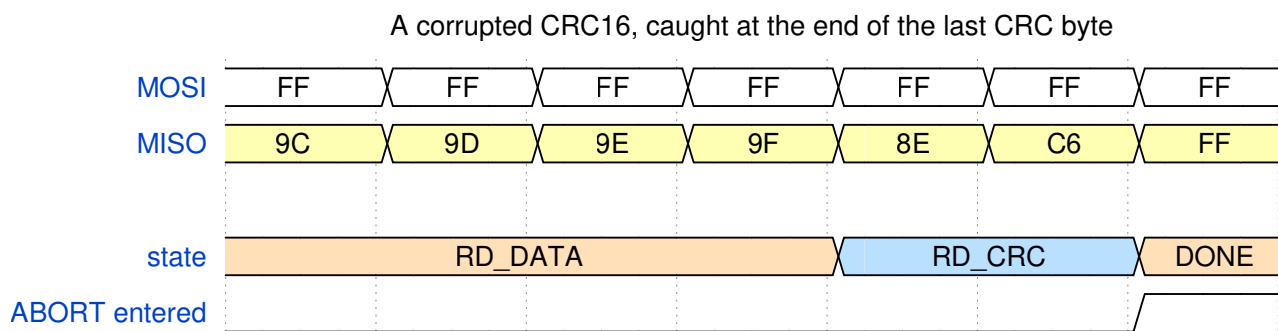
5.4 The end of a block write



The data-response token is the trap here. Its shape is `xxx0sss1` — **five bits of meaning in eight** — so mask with `0x1F` before comparing. Comparing the whole byte against a constant is the usual way to get this wrong.

Here it is `0x05`: `sss = 010`, data accepted. The card then holds `MISO` low while it programs the block, which the core reports as `CARD_BUSY`.

5.5 A CRC16 that does not match



Nothing on the wire distinguishes a corrupt block from a good one until the check is done. The core folds the last byte into the running CRC16 combinationally and compares against zero, so the mismatch is known at the end of the final CRC byte rather than a byte later.

`ABORT` needs a row of its own because it lasts a cycle or two — it raises `dma_abort`, then routes to `DONE` — so a state row sampled once per byte never lands on it.

There is no `irq` row because it would be flat high: `DMA_DONE` raised the pin earlier in the data phase. The pin says something happened; only `IRQ_STATUS` says what.

6. Regenerating the figures

```
python3 doc/tools/diagrams/build_figures.py # block diagrams, via Graphviz
./verification/capture.sh                  # record verification/wave.vcd
python3 doc/tools/waveforms/mkwaves.py    # timing figures, via WaveDrom
python3 doc/tools/build_pdf.py all        # typeset both documents
./verification/check_figures.sh           # confirm the tracked SVGs match
```

The block diagrams are Graphviz sources, so the layout is computed rather than placed by hand. The timing figures are generated from the VCD: change the RTL and either the figure changes with it, or `mkwaves.py` stops finding the token or state it is looking for and says which.

Requirements: Graphviz for the block diagrams; Verilator for the capture; Node plus `npm install wavedrom onml` in `doc/tools/waveforms` for the timing figures. The tracked SVGs mean none of that is needed just to read the documents.

Revision history

Version	Change
v1.1	Block diagrams moved to Graphviz, timing figures to WaveDrom. Register-map figure dropped — it was a table drawn as a picture.
v1.0	First issue.